
UNIT 1: NUMBER SYSTEMS

CHAPTER 1: REAL NUMBERS

1. Fundamental Theorem of Arithmetic

Statement:

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.

Example:

- $12 = 2 \times 2 \times 3 = 2^2 \times 3$
- $84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$
- $360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 2^3 \times 3^2 \times 5$

Application:

- Used to find HCF and LCM of two or more positive integers.

HCF (Highest Common Factor):

- Product of the smallest power of each common prime factor.

LCM (Least Common Multiple):

- Product of the greatest power of each prime factor.

Relationship between HCF and LCM:

For any two positive integers a and b:

- $a \times b = \text{HCF}(a, b) \times \text{LCM}(a, b)$

Example:

For 12 and 18:

- $12 = 2^2 \times 3$
- $18 = 2 \times 3^2$
- $\text{HCF} = 2 \times 3 = 6$
- $\text{LCM} = 2^2 \times 3^2 = 36$
- $12 \times 18 = 216$
- $6 \times 36 = 216$

****Important:****

This formula holds true only for two numbers. For three numbers, the relationship is:

- $a \times b \times c = \text{HCF}(a, b, c) \times \text{LCM}(a, b, c)$
- This is not generally true. The correct relationship is:
- $\text{HCF}(a, b, c) \times \text{LCM}(a, b, c) \neq a \times b \times c$

****Applications of HCF and LCM:****

- HCF is used when we need to divide things into smaller equal parts.
- LCM is used when we need to find the time when events occur simultaneously.

2. Prime Factorisation Method for Finding HCF and LCM

****Steps:****

1. Find the prime factors of each number.
2. For HCF, multiply the common prime factors with the smallest exponent.
3. For LCM, multiply all prime factors with the greatest exponent.

****Example:****

Find HCF and LCM of 24, 36, and 40.

- $24 = 2^3 \times 3$
- $36 = 2^2 \times 3^2$
- $40 = 2^3 \times 5$
- $\text{HCF} = 2^2 = 4$
- $\text{LCM} = 2^3 \times 3^2 \times 5 = 8 \times 9 \times 5 = 360$

3. Irrational Numbers

****Irrational Number:****

A number that cannot be expressed in the form p/q , where p and q are integers and $q \neq 0$.

- Examples: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , $3 + 2\sqrt{5}$

****Proof of Irrationality:****

To prove that $\sqrt{2}$ is irrational:

****Method:****

1. Assume that $\sqrt{2}$ is rational.
2. Then $\sqrt{2} = p/q$, where p and q are coprime integers ($\text{HCF} = 1$) and $q \neq 0$.
3. Squaring both sides: $2 = p^2/q^2 \rightarrow p^2 = 2q^2$
4. This implies p^2 is even, so p is even.

5. Let $p = 2k$. Then $(2k)^2 = 2q^2 \rightarrow 4k^2 = 2q^2 \rightarrow 2k^2 = q^2$
6. This implies q^2 is even, so q is even.
7. Thus p and q are both even, contradicting that they are coprime.
8. Therefore, our assumption is wrong. $\sqrt{2}$ is irrational.

****Similarly, we can prove that:****

- $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, etc., are irrational.
- $3 + 2\sqrt{5}$ is irrational.
- $1/\sqrt{2}$ is irrational.

****Important:****

- The sum of a rational and an irrational number is irrational.
- The product of a non-zero rational and an irrational number is irrational.

4. Terminating and Non-Terminating Decimals

****Terminating Decimals:****

- Decimals that come to an end.
- Example: 0.5, 1.25, 3.75

****Non-Terminating Repeating Decimals:****

- Decimals that repeat a pattern.
- Example: 0.333..., 0.142857142857...

****Rational Numbers and Decimals:****

- A rational number has a terminating decimal expansion if and only if the denominator (in its simplest form) has prime factors only 2 or 5.
- Otherwise, it has a non-terminating repeating decimal expansion.

****Example:****

- $7/8 = 0.875$ (terminating, denominator $8 = 2^3$)
- $3/25 = 0.12$ (terminating, denominator $25 = 5^2$)
- $1/6 = 0.1666...$ (non-terminating, denominator $6 = 2 \times 3$)
- $2/7 = 0.285714285714...$ (non-terminating, denominator 7)

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. State the Fundamental Theorem of Arithmetic.
2. Write the prime factorisation of 72.

3. What is the HCF of two prime numbers?
4. Is π a rational or irrational number?
5. What is the LCM of 12 and 18?

****Short Answer (2-3 marks):****

1. Find the HCF and LCM of 24 and 36 using the prime factorisation method.
2. Prove that $\sqrt{3}$ is irrational.
3. Show that $2 + \sqrt{5}$ is an irrational number.
4. Find the HCF and LCM of 96 and 404 and verify that $\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$.

****Long Answer (4-5 marks):****

1. Find the LCM and HCF of the following numbers using the prime factorisation method:
 - 24, 36, 40
 - 16, 24, 36
2. Prove that $3 + 2\sqrt{5}$ is irrational.
3. Explain the condition for a rational number to have a terminating decimal expansion. Give examples.

****Case-Based Question:****

Read the following and answer:

"Raju and Ravi are two friends. Raju visits a library every 12 days, and Ravi visits it every 18 days. They both visited the library today."

- (i) After how many days will they visit the library together again?
- (ii) What concept is used to find this?
- (iii) What is the HCF of 12 and 18?
- (iv) What is the prime factorisation of 18?

UNIT 2: ALGEBRA

CHAPTER 2: POLYNOMIALS

1. Definition and Types of Polynomials

****Polynomial:****

An algebraic expression of the form:

$$- p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

- Where $a_n, a_{n-1}, \dots, a_1, a_0$ are real numbers and $a_n \neq 0$.
- The exponents of x are non-negative integers.

Examples:

- $3x^2 + 2x + 1$ (Polynomial)
- $x^3 - 4x + 7$ (Polynomial)
- $2x^2 + 3/x$ (Not a polynomial, exponent of x is -1)
- $\sqrt{x} + 1$ (Not a polynomial, exponent of x is $1/2$)

Degree of a Polynomial:

The highest power of x in the polynomial.

- **Linear:** Degree 1 (Example: $ax + b$)
- **Quadratic:** Degree 2 (Example: $ax^2 + bx + c$)
- **Cubic:** Degree 3 (Example: $ax^3 + bx^2 + cx + d$)
- **Constant:** Degree 0 (Example: 5)

Zero Polynomial:

A polynomial with all coefficients equal to zero.

- Example: $0x^2 + 0x + 0$
- Degree of zero polynomial is not defined.

Types based on Number of Terms:

- **Monomial:** One term (Example: $3x^2$)
- **Binomial:** Two terms (Example: $2x + 1$)
- **Trinomial:** Three terms (Example: $x^2 + 2x + 1$)

Coefficients:

- In $p(x) = 3x^2 + 2x + 1$
- Coefficient of $x^2 = 3$
- Coefficient of $x = 2$
- Constant term = 1

2. Zeroes of a Polynomial

Zero of a Polynomial:

A value of x for which $p(x) = 0$.

Geometric Meaning:

- The zeroes of a polynomial are the x -coordinates of the points where the graph of $y = p(x)$ intersects the x -axis.
- For a linear polynomial $ax + b$, the zero is $-b/a$.
- For a quadratic polynomial $ax^2 + bx + c$:

- If the graph intersects the x-axis at two points, it has two zeroes.
- If it touches the x-axis at one point, it has one zero (repeated).
- If it does not intersect the x-axis, it has no real zeroes.

****Number of Zeroes:****

- A polynomial of degree n has at most n zeroes.

****Example:****

- For $p(x) = x^2 - 5x + 6$:
 - Zeroes are 2 and 3.
 - $p(2) = 4 - 10 + 6 = 0$
 - $p(3) = 9 - 15 + 6 = 0$

3. Relationship between Zeroes and Coefficients (Quadratic Polynomials)

For a quadratic polynomial $p(x) = ax^2 + bx + c$, with zeroes α and β :

- Sum of zeroes: $\alpha + \beta = -b/a$
- Product of zeroes: $\alpha \times \beta = c/a$

****Example:****

- For $p(x) = 2x^2 - 7x + 6$:
 - $a = 2$, $b = -7$, $c = 6$
 - $\alpha + \beta = -(-7)/2 = 7/2$
 - $\alpha \times \beta = 6/2 = 3$

****Verification:****

- Zeroes are $3/2$ and 2 .
- $\alpha + \beta = 3/2 + 2 = 7/2$
- $\alpha \times \beta = (3/2) \times 2 = 3$

****Forming a Quadratic Polynomial:****

If α and β are the zeroes:

- $p(x) = k[x^2 - (\alpha + \beta)x + \alpha\beta]$
- Where k is a constant.

****Example:****

If zeroes are 2 and 3:

- Sum = 5
- Product = 6
- $p(x) = k(x^2 - 5x + 6)$

4. Division Algorithm for Polynomials

****Statement:****

If $p(x)$ and $g(x)$ are two polynomials with $g(x) \neq 0$, then we can find polynomials $q(x)$ and $r(x)$ such that:

- $p(x) = g(x) \times q(x) + r(x)$
- Where $r(x) = 0$ or degree of $r(x) < \text{degree of } g(x)$.

****Example:****

Divide $p(x) = x^3 - 3x^2 + x + 2$ by $g(x) = x - 2$.

- Quotient $q(x) = x^2 - x - 1$
- Remainder $r(x) = 0$

****Steps for Division:****

1. Arrange the polynomials in decreasing order of powers.
2. Divide the first term of the dividend by the first term of the divisor.
3. Multiply the divisor by this term and subtract from the dividend.
4. Repeat the process with the remainder.

5. Algebraic Identities

****Important Identities:****

- $(x + y)^2 = x^2 + 2xy + y^2$
- $(x - y)^2 = x^2 - 2xy + y^2$
- $(x + y)(x - y) = x^2 - y^2$
- $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$
- $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$
- $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$
- $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

****Special Case:****

If $x + y + z = 0$, then $x^3 + y^3 + z^3 = 3xyz$

****Example:****

$$- 8x^3 + 27y^3 = (2x)^3 + (3y)^3 = (2x + 3y)(4x^2 - 6xy + 9y^2)$$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. What is the degree of polynomial $3x^2 + 5x + 1$?
2. Find the sum of zeroes of $x^2 - 5x + 6$.
3. What is the product of zeroes of $2x^2 - 3x + 4$?
4. Is 2 a zero of $x^2 - 4x + 4$? Justify your answer.
5. Write the identity for $(x + y)^3$.

****Short Answer (2-3 marks):****

1. Find the zeroes of $x^2 - 7x + 12$ and verify the relationship between zeroes and coefficients.
2. If α and β are zeroes of $x^2 - 5x + 6$, find the value of $\alpha^2 + \beta^2$.
3. Find a quadratic polynomial whose zeroes are 2 and -3.
4. Divide $x^3 - 4x^2 + x + 6$ by $x - 2$.

****Long Answer (4-5 marks):****

1. Find the zeroes of $2x^2 - 7x + 3$ and verify the relationship between zeroes and coefficients.
2. If α and β are zeroes of $x^2 + 6x + 5$, find the value of $1/\alpha + 1/\beta$.
3. Divide $3x^3 + x^2 - 7x + 2$ by $x + 2$ and verify the division algorithm.

****Case-Based Question:****

Read the following and answer:

"A quadratic polynomial has zeroes 3 and 4."

- (i) Write the polynomial in terms of x .
- (ii) What is the sum of the zeroes?
- (iii) What is the product of the zeroes?
- (iv) What is the degree of this polynomial?

CHAPTER 3: PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

1. Linear Equations in Two Variables

****General Form:****

- $ax + by + c = 0$
- Where a, b, c are real numbers, and a and b are not both zero.

****Solution:****

- A pair of values (x, y) that satisfies the equation.
- A linear equation in two variables has infinitely many solutions.

****Example:****

- $2x + 3y = 6$
- Solutions: $(0, 2)$, $(3, 0)$, $(1, 4/3)$, etc.

****Graph:****

- The graph of a linear equation in two variables is a straight line.

2. Pair of Linear Equations in Two Variables

****General Form:****

- $a_1x + b_1y + c_1 = 0$
- $a_2x + b_2y + c_2 = 0$

****Solutions:****

- A pair of values (x, y) that satisfies both equations.

****Types of Solutions:****

****A. Consistent and Dependent:****

- Infinite solutions.
- Lines coincide.
- Condition: $a_1/a_2 = b_1/b_2 = c_1/c_2$

****Example:****

- $2x + 3y = 6$ and $4x + 6y = 12$

****B. Consistent and Independent:****

- Unique solution.
- Lines intersect at one point.
- Condition: $a_1/a_2 \neq b_1/b_2$

****Example:****

- $2x + 3y = 6$ and $3x + 2y = 12$

****C. Inconsistent:****

- No solution.
- Lines are parallel.
- Condition: $a_1/a_2 = b_1/b_2 \neq c_1/c_2$

****Example:****

- $2x + 3y = 6$ and $4x + 6y = 24$

3. Graphical Method of Solution

Steps:

1. Draw the graph of both equations.
2. Find the point where the two lines intersect.
3. The coordinates of the point of intersection give the solution.

Example:

Solve: $2x + 3y = 6$ and $x + 2y = 4$

Step 1:

- For $2x + 3y = 6$:

- $x = 0 \rightarrow y = 2 \rightarrow (0, 2)$
- $y = 0 \rightarrow x = 3 \rightarrow (3, 0)$

- For $x + 2y = 4$:

- $x = 0 \rightarrow y = 2 \rightarrow (0, 2)$
- $y = 0 \rightarrow x = 4 \rightarrow (4, 0)$

Step 2:

- The lines intersect at $(0, 2)$.
- Solution: $x = 0, y = 2$

4. Algebraic Methods of Solution

A. Substitution Method:

Steps:

1. Express one variable in terms of the other from one equation.
2. Substitute this value in the second equation.
3. Solve for one variable.
4. Find the other variable.

Example:

Solve: $2x + 3y = 6$ and $x + 2y = 4$

From the second equation:

$$- x = 4 - 2y$$

Substitute in the first:

$$\begin{aligned}
 & -2(4 - 2y) + 3y = 6 \\
 & -8 - 4y + 3y = 6 \\
 & -8 - y = 6 \\
 & -y = 2
 \end{aligned}$$

Substitute $y = 2$ in $x = 4 - 2y$:

$$-x = 4 - 4 = 0$$

Solution: $x = 0, y = 2$

****B. Elimination Method:****

****Steps:****

1. Multiply one or both equations by suitable numbers to make the coefficients of one variable equal.
2. Add or subtract the equations to eliminate that variable.
3. Solve for the remaining variable.
4. Substitute to find the other variable.

****Example:****

Solve: $2x + 3y = 6$ and $x + 2y = 4$

Multiply the second equation by 2:

$$-2x + 4y = 8$$

Subtract the first equation from this:

$$\begin{aligned}
 & -(2x + 4y) - (2x + 3y) = 8 - 6 \\
 & -y = 2
 \end{aligned}$$

Substitute $y = 2$ in $x + 2y = 4$:

$$\begin{aligned}
 & -x + 4 = 4 \\
 & -x = 0
 \end{aligned}$$

Solution: $x = 0, y = 2$

****C. Cross-Multiplication Method:****

For equations:

$$\begin{aligned}
 & -a_1x + b_1y + c_1 = 0 \\
 & -a_2x + b_2y + c_2 = 0
 \end{aligned}$$

The solution is:

$$-x/(b_1c_2 - b_2c_1) = y/(c_1a_2 - c_2a_1) = 1/(a_1b_2 - a_2b_1)$$

5. Word Problems

****Types of Problems:****

1. ****Problems on Age:****

- Form equations based on the given conditions.

2. ****Problems on Numbers:****

- Form equations for two-digit numbers, sum of digits, etc.

3. ****Problems on Rates:****

- Speed, distance, time.
- Work and time.

4. ****Problems on Money:****

- Amounts, interest, investments.

****Example:****

The sum of two numbers is 15 and their difference is 3. Find the numbers.

Let the numbers be x and y.

$$-x + y = 15$$

$$-x - y = 3$$

Adding:

$$-2x = 18$$

$$-x = 9$$

Substitute:

$$-9 + y = 15$$

$$-y = 6$$

Numbers are 9 and 6.

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. What is the general form of a linear equation in two variables?
2. For what value of k will the equations $x + 2y = 3$ and $2x + 4y = k$ have infinitely many solutions?

3. What is the condition for two linear equations to have a unique solution?
4. How many solutions does a pair of linear equations have if the lines are parallel?

****Short Answer (2-3 marks):****

1. Solve graphically: $2x + y = 6$ and $3x + 2y = 5$.
2. Solve by substitution method: $x + y = 5$ and $2x - y = 4$.
3. Solve by elimination method: $3x - y = 7$ and $x + 2y = 9$.
4. Check whether the following pair of equations is consistent or inconsistent: $2x + 3y = 8$ and $4x + 6y = 16$.

****Long Answer (4-5 marks):****

1. Solve the following pair of equations by the elimination method:
- $2x + 3y = 11$ and $3x - 2y = 10$
2. Solve graphically: $x - y = 1$ and $2x + y = 8$. Find the area of the triangle formed by these lines and the x-axis.
3. The sum of two numbers is 45. The difference of their reciprocals is $1/6$. Find the numbers.

****Case-Based Question:****

Read the following and answer:

"A man is 10 years older than his son. In 5 years, the sum of their ages will be 60 years."

- (i) Write the equations for the given situation.
- (ii) How old is the son now?
- (iii) How old is the man now?
- (iv) What method would you use to solve these equations?

CHAPTER 4: QUADRATIC EQUATIONS

1. Quadratic Equations

****Standard Form:****

- $ax^2 + bx + c = 0$, where $a \neq 0$
- a, b, c are real numbers.

****Examples:****

- $2x^2 + 3x - 5 = 0$ (a = 2, b = 3, c = -5)
- $x^2 - 4x = 0$ (a = 1, b = -4, c = 0)
- $3x^2 + 2 = 0$ (a = 3, b = 0, c = 2)

****Roots of a Quadratic Equation:****

- The values of x that satisfy the equation.

2. Methods of Solving Quadratic Equations

A. Factorisation Method:

Steps:

1. Factorise the quadratic expression.
2. Set each factor equal to zero.
3. Solve for x.

Example:

Solve: $x^2 - 5x + 6 = 0$

$$- x^2 - 5x + 6 = (x - 2)(x - 3)$$

$$- x - 2 = 0 \rightarrow x = 2$$

$$- x - 3 = 0 \rightarrow x = 3$$

Example:

Solve: $2x^2 - 7x + 3 = 0$

$$- 2x^2 - 7x + 3 = (2x - 1)(x - 3)$$

$$- 2x - 1 = 0 \rightarrow x = 1/2$$

$$- x - 3 = 0 \rightarrow x = 3$$

B. Quadratic Formula:

Formula:

$$- x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$$

Example:

Solve: $2x^2 + 3x - 5 = 0$

$$- a = 2, b = 3, c = -5$$

$$- b^2 - 4ac = 9 - 4(2)(-5) = 9 + 40 = 49$$

$$- x = [-3 \pm \sqrt{49}] / 4 = [-3 \pm 7] / 4$$

$$- x = 4/4 = 1$$

$$- x = -10/4 = -5/2$$

C. Completing the Square:

****Steps:****

1. Write the equation in the form $ax^2 + bx = -c$.
2. Divide by a if necessary.
3. Add $(b/2a)^2$ to both sides.
4. Write the left side as a perfect square.
5. Solve for x .

****Example:****

Solve: $x^2 + 4x - 5 = 0$

- $x^2 + 4x = 5$
- Add $(4/2)^2 = 4$ to both sides:
- $x^2 + 4x + 4 = 9$
- $(x + 2)^2 = 9$
- $x + 2 = \pm 3$
- $x = 1$ or $x = -5$

3. Nature of Roots

****Discriminant (D):****

- $D = b^2 - 4ac$

****Nature of Roots:****

- **** $D > 0$:** Two distinct real roots.
- **** $D = 0$:** Two equal real roots (one repeated root).
- **** $D < 0$:** No real roots (complex roots).

****Example:****

For $2x^2 + 3x - 5 = 0$:

- $D = 9 - 4(2)(-5) = 49 > 0 \rightarrow$ Two distinct real roots.

For $x^2 - 4x + 4 = 0$:

- $D = 16 - 4(1)(4) = 0 \rightarrow$ One repeated root ($x = 2$).

For $x^2 + 4x + 5 = 0$:

- $D = 16 - 4(1)(5) = -4 < 0 \rightarrow$ No real roots.

4. Word Problems

****Types of Problems:****

1. ****Problems on Numbers:****

- Let the number be x .
- 2. **Problems on Ages:**
 - Form quadratic equations based on age conditions.
- 3. **Problems on Areas:**
 - Areas of rectangles, squares, circles, etc.
- 4. **Problems on Speed, Distance, Time:**
 - Use the formula: Distance = Speed $\times$ Time.
- 5. **Problems on Profit and Loss:**
 - Cost price, selling price, profit, loss.

Example:

The product of two consecutive positive integers is 72. Find the numbers.

Let the numbers be x and $x + 1$.

- $x(x + 1) = 72$
- $x^2 + x - 72 = 0$
- $x^2 + 9x - 8x - 72 = 0$
- $x(x + 9) - 8(x + 9) = 0$
- $(x - 8)(x + 9) = 0$
- $x = 8$ or $x = -9$
- Since numbers are positive, $x = 8$.
- Numbers are 8 and 9.

Example:

The sum of the areas of two squares is 468 m^2 . If the difference of their perimeters is 24 m, find the sides.

Let sides be x and y .

- $x^2 + y^2 = 468$
- $4x - 4y = 24 \rightarrow x - y = 6 \rightarrow x = y + 6$

Substitute:

- $(y + 6)^2 + y^2 = 468$
- $y^2 + 12y + 36 + y^2 = 468$
- $2y^2 + 12y - 432 = 0$
- $y^2 + 6y - 216 = 0$
- $y^2 + 18y - 12y - 216 = 0$
- $(y + 18)(y - 12) = 0$
- $y = 12$ or $y = -18$ (reject negative)
- $y = 12, x = 18$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the standard form of a quadratic equation.
2. What is the discriminant of a quadratic equation?
3. For what value of k does $x^2 + 4x + k = 0$ have equal roots?
4. Find the nature of roots of $x^2 - 6x + 9 = 0$.
5. What is the sum of roots of $2x^2 - 5x + 3 = 0$?

****Short Answer (2-3 marks):****

1. Solve: $x^2 - 10x + 21 = 0$ by factorisation.
2. Solve: $2x^2 + 3x - 5 = 0$ using the quadratic formula.
3. Find the nature of roots of $2x^2 - 5x + 3 = 0$.
4. If one root of $x^2 - 6x + k = 0$ is 2, find the value of k and the other root.

****Long Answer (4-5 marks):****

1. Solve: $2x^2 - 7x + 3 = 0$ by completing the square.
2. The sum of two numbers is 15 and the sum of their squares is 113. Find the numbers.
3. A train travels 480 km at a uniform speed. If the speed had been 8 km/h less, it would have taken 3 hours more to cover the same distance. Find the speed of the train.

****Case-Based Question:****

Read the following and answer:

"The product of two consecutive positive integers is 156."

- (i) Write the quadratic equation for this situation.
- (ii) What are the numbers?
- (iii) What method would you use to solve this equation?
- (iv) What is the sum of the numbers?

CHAPTER 5: ARITHMETIC PROGRESSIONS

1. Arithmetic Progression (AP)

****Definition:****

A sequence in which the difference between any two consecutive terms is constant.

****Common Difference (d):****

$$-d = a_2 - a_1 = a_3 - a_2 = \dots$$

****Example:****

- 2, 5, 8, 11, ... ($d = 3$)
- 10, 7, 4, 1, ... ($d = -3$)
- 3, 3, 3, 3, ... ($d = 0$)

****General Term:****

- $a, a + d, a + 2d, a + 3d, \dots$

2. nth Term of an AP

****Formula:****

- $a_n = a + (n - 1)d$
- Where:
 - a = first term
 - d = common difference
 - n = number of terms

****Example:****

Find the 20th term of the AP: 2, 5, 8, 11, ...

- $a = 2, d = 3, n = 20$
- $a_{20} = 2 + (20 - 1)3 = 2 + 57 = 59$

****Example:****

How many terms are there in the AP: 3, 7, 11, ..., 47?

- $a = 3, d = 4, a_n = 47$
- $47 = 3 + (n - 1)4$
- $47 = 3 + 4n - 4$
- $47 = 4n - 1$
- $4n = 48$
- $n = 12$

3. Sum of First n Terms of an AP

****Formula:****

- $S_n = \frac{n}{2}[2a + (n - 1)d]$
- Or $S_n = \frac{n}{2}(a + l)$, where l is the last term.

****Example:****

Find the sum of the first 20 terms of the AP: 2, 5, 8, 11, ...

- $a = 2, d = 3, n = 20$
- $S_{20} = \frac{20}{2}[2(2) + (20 - 1)3] = 10[4 + 57] = 10 \times 61 = 610$

****Example:****

Find the sum of the AP: 3, 7, 11, ..., 47.

- $a = 3, l = 47, n = 12$

- $S_{12} = 12/2(3 + 47) = 6 \times 50 = 300$

****Example:****

The sum of the first n terms of an AP is given by $S_n = n^2 - 2n$. Find the AP.

- $S_1 = 1^2 - 2(1) = -1 \rightarrow a = -1$

- $S_2 = 2^2 - 2(2) = 0 \rightarrow a_2 = S_2 - S_1 = 0 - (-1) = 1$

- $d = a_2 - a_1 = 1 - (-1) = 2$

- AP: -1, 1, 3, 5, ...

4. Application of AP in Daily Life

****Types of Problems:****

1. ****Problems on Sums:****

- The sum of n terms is given; find the AP or the terms.

2. ****Problems on Ages:****

- Ages in AP.

3. ****Problems on Money:****

- Interest, instalments, etc.

4. ****Problems on Time:****

- Time intervals in AP.

****Example:****

A manufacturer produces 100 units in the first year and increases production by 20 units every year. Find the total production in 10 years.

- $a = 100, d = 20, n = 10$

- $S_{10} = 10/2[2(100) + (10 - 1)20] = 5[200 + 180] = 5 \times 380 = 1900$ units.

****Example:****

The sum of three numbers in AP is 21 and their product is 231. Find the numbers.

- Let numbers be $a - d, a, a + d$.

- $(a - d) + a + (a + d) = 21 \rightarrow 3a = 21 \rightarrow a = 7$

- $(a - d)(a)(a + d) = 231 \rightarrow (7 - d)(7)(7 + d) = 231$

- $7(49 - d^2) = 231$

- $49 - d^2 = 33$

- $d^2 = 16$

- $d = \pm 4$

Numbers: 3, 7, 11 or 11, 7, 3

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the formula for the n th term of an AP.
2. What is the common difference of the AP: 3, 7, 11, 15, ...?
3. What is the sum of the first n terms of an AP?
4. Find the 10th term of the AP: 2, 5, 8, ...
5. What is the value of d if $a = 5$ and $a_3 = 11$?

****Short Answer (2-3 marks):****

1. Find the 15th term of the AP: 5, 9, 13, 17, ...
2. How many terms are there in the AP: 2, 5, 8, ..., 65?
3. Find the sum of the first 12 terms of the AP: 4, 9, 14, 19, ...
4. Find the AP whose sum of first n terms is $3n^2 + 2n$.

****Long Answer (4-5 marks):****

1. The sum of first 7 terms of an AP is 49 and the sum of first 17 terms is 289. Find the AP.
2. The 5th and 13th terms of an AP are 17 and 33 respectively. Find the AP.
3. A man starts repaying a loan with the first instalment of Rs. 500. He increases his instalment by Rs. 25 every month. What will be the total amount he pays in 10 months?

****Case-Based Question:****

Read the following and answer:

"The sum of the first 5 terms of an AP is 60 and the sum of the first 10 terms is 210."

- (i) Find the first term and common difference of the AP.
- (ii) What is the 10th term of the AP?
- (iii) What is the sum of the 6th to 10th terms?
- (iv) Write the AP.

UNIT 3: COORDINATE GEOMETRY

CHAPTER 6: COORDINATE GEOMETRY

1. Coordinate Plane

Cartesian Plane:

- Two perpendicular lines (x-axis and y-axis) intersect at the origin.
- The plane is divided into four quadrants.

Coordinates of a Point:

- (x, y)
- x = distance from the y-axis (abscissa)
- y = distance from the x-axis (ordinate)

Quadrants:

- **Quadrant I:** (+, +)
- **Quadrant II:** (-, +)
- **Quadrant III:** (-, -)
- **Quadrant IV:** (+, -)

2. Distance Formula

Statement:

The distance between two points A(x₁, y₁) and B(x₂, y₂) is:

$$- AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Derivation:

Using the Pythagorean theorem.

Example:

Find the distance between A(2, 3) and B(5, 7).

$$- AB = \sqrt{(5 - 2)^2 + (7 - 3)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Example:

Find the distance between P(-3, 4) and Q(2, -4).

$$- PQ = \sqrt{(2 - (-3))^2 + (-4 - 4)^2} = \sqrt{5^2 + (-8)^2} = \sqrt{25 + 64} = \sqrt{89}$$

Applications:

- Finding the distance between two points.
- Checking if three points are collinear.
- Proving properties of triangles and quadrilaterals.

3. Section Formula

****Statement:****

The coordinates of a point P(x, y) dividing the line segment joining A(x₁, y₁) and B(x₂, y₂) in the ratio m₁ : m₂ are:

$$- x = (m_1 x_2 + m_2 x_1) / (m_1 + m_2)$$

$$- y = (m_1 y_2 + m_2 y_1) / (m_1 + m_2)$$

****Internal Division:****

- The point lies between A and B.

****Midpoint Formula:****

If m₁ = m₂ = 1:

$$- x = (x_1 + x_2) / 2$$

$$- y = (y_1 + y_2) / 2$$

****Example:****

Find the coordinates of the point dividing A(2, 3) and B(6, 7) in the ratio 3 : 2.

$$- x = (3 \times 6 + 2 \times 2) / (3 + 2) = (18 + 4) / 5 = 22/5$$

$$- y = (3 \times 7 + 2 \times 3) / 5 = (21 + 6) / 5 = 27/5$$

****Applications:****

- Finding the midpoint of a line segment.
- Finding the centroid of a triangle.
- Finding a point on a line segment.

4. Area of a Triangle

****Formula:****

Area of triangle with vertices A(x₁, y₁), B(x₂, y₂), and C(x₃, y₃):

$$- \text{Area} = 1/2 |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

****Alternative Formula:****

$$- \text{Area} = 1/2 |(x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2))|$$

****Collinearity:****

- Three points are collinear if the area of the triangle formed by them is zero.

****Example:****

Find the area of triangle with vertices A(1, 2), B(3, 4), and C(5, 1).

$$- \text{Area} = 1/2 |1(4 - 1) + 3(1 - 2) + 5(2 - 4)|$$

$$- = 1/2 |1(3) + 3(-1) + 5(-2)|$$

$$- = 1/2 |3 - 3 - 10| = 1/2 |-10| = 5 \text{ square units}$$

****Example:****

Check if A(1, 2), B(3, 4), and C(5, 6) are collinear.

$$\text{- Area} = \frac{1}{2} |1(4 - 6) + 3(6 - 2) + 5(2 - 4)|$$

$$\text{-} = \frac{1}{2} |1(-2) + 3(4) + 5(-2)|$$

$$\text{-} = \frac{1}{2} |-2 + 12 - 10| = \frac{1}{2} |0| = 0$$

- Therefore, the points are collinear.

5. Centroid of a Triangle

****Formula:****

The centroid G of triangle with vertices A(x_1 , y_1), B(x_2 , y_2), and C(x_3 , y_3):

$$\text{- } G = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$$

****Example:****

Find the centroid of triangle with vertices A(1, 2), B(3, 4), and C(5, 6).

$$\text{- } G = ((1 + 3 + 5)/3, (2 + 4 + 6)/3) = (9/3, 12/3) = (3, 4)$$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the distance formula.
2. Write the section formula for internal division.
3. Write the formula for the area of a triangle.
4. What is the midpoint formula?
5. How many coordinates does a point in a plane have?

****Short Answer (2-3 marks):****

1. Find the distance between A(2, 3) and B(4, 7).
2. Find the midpoint of the line segment joining A(2, 5) and B(6, 9).
3. Find the coordinates of the point dividing A(2, 3) and B(6, 9) in the ratio 1 : 3.
4. Find the area of the triangle with vertices A(1, 2), B(3, 4), and C(5, 1).

****Long Answer (4-5 marks):****

1. Find the coordinates of the point on the y-axis which is equidistant from A(2, 3) and B(4, 5).
2. Prove that A(2, 3), B(4, 5), and C(6, 7) are collinear.
3. Find the area of the triangle whose vertices are (3, 1), (5, 3), and (1, 3). Also find the centroid of the triangle.

****Case-Based Question:****

Read the following and answer:

"A point P divides the line segment joining A(2, 3) and B(6, 7) in the ratio 2 : 3."

- (i) Find the coordinates of P.
- (ii) What is the formula used to find P?
- (iii) What would be the coordinates of the midpoint of AB?
- (iv) What is the distance between A and B?

UNIT 4: GEOMETRY

CHAPTER 7: TRIANGLES

1. Similarity of Triangles

Definition:

Two triangles are similar if:

1. Their corresponding angles are equal.
2. Their corresponding sides are proportional.

Symbol:

- $\triangle ABC \sim \triangle DEF$

Conditions for Similarity:

A. AAA (Angle-Angle-Angle):

- If all three corresponding angles are equal.

B. SSS (Side-Side-Side):

- If all three corresponding sides are proportional.

C. SAS (Side-Angle-Side):

- If two corresponding sides are proportional and the included angle is equal.

2. Basic Proportionality Theorem (BPT) or Thales Theorem

Statement:

If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.

****Proof:****

In $\triangle ABC$, $DE \parallel BC$, intersecting AB at D and AC at E .

- To prove: $AD/DB = AE/EC$

****Construction:****

- Join BE and CD .

- Draw $EF \parallel AB$ and $DG \parallel AC$.

****Steps:****

1. Area of $\triangle ADE = \frac{1}{2} \times AD \times EF$

2. Area of $\triangle BDE = \frac{1}{2} \times DB \times EF$

3. Area of $\triangle ADE$ / Area of $\triangle BDE = AD/DB$

4. Area of $\triangle ADE = \frac{1}{2} \times AE \times DG$

5. Area of $\triangle CDE = \frac{1}{2} \times EC \times DG$

6. Area of $\triangle ADE$ / Area of $\triangle CDE = AE/EC$

7. Since $\triangle BDE$ and $\triangle CDE$ are on the same base DE and between the same parallels DE and BC , their areas are equal.

8. Therefore, $AD/DB = AE/EC$

3. Converse of BPT

****Statement:****

If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.

4. Pythagoras Theorem

****Statement:****

In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

****Proof:****

In $\triangle ABC$, $\angle B = 90^\circ$. To prove: $AC^2 = AB^2 + BC^2$

****Construction:****

- Draw $BD \perp AC$.

****Steps:****

1. In $\triangle ABC$ and $\triangle ABD$:

- $\angle A$ is common.
- $\angle ABC = \angle ADB = 90^\circ$
- $\triangle ABC \sim \triangle ABD$ (AA similarity)
- $AC/AB = AB/AD \rightarrow AB^2 = AC \times AD \dots (1)$

2. In $\triangle ABC$ and $\triangle BDC$:

- $\angle C$ is common.
- $\angle ABC = \angle BDC = 90^\circ$
- $\triangle ABC \sim \triangle BDC$ (AA similarity)
- $AC/BC = BC/DC \rightarrow BC^2 = AC \times DC \dots (2)$

3. Adding (1) and (2):

$$- AB^2 + BC^2 = AC(AD + DC) = AC \times AC = AC^2$$

5. Converse of Pythagoras Theorem

****Statement:****

If the square of one side of a triangle is equal to the sum of the squares of the other two sides, then the triangle is right-angled.

6. Important Theorems

****1.** The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. ******

****2.** The corresponding medians of similar triangles are proportional to the corresponding sides. ******

****3.** The corresponding altitudes of similar triangles are proportional to the corresponding sides. ******

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the Basic Proportionality Theorem.
2. What is the condition for two triangles to be similar?
3. State the Pythagoras Theorem.
4. In a right-angled triangle, if the two legs are 3 cm and 4 cm, find the hypotenuse.
5. Write the converse of the BPT.

****Short Answer (2-3 marks):****

1. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
2. In $\triangle ABC$, $DE \parallel BC$. If $AD = 2$ cm, $DB = 3$ cm, and $AE = 4$ cm, find EC .
3. Prove that in a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.
4. If a ladder of length 13 m is placed against a wall such that its foot is 5 m away from the wall, find the height of the wall.

****Long Answer (4-5 marks):****

1. Prove the Basic Proportionality Theorem.
2. Prove the Pythagoras Theorem using the concept of similar triangles.
3. In a right-angled triangle ABC , with $\angle B = 90^\circ$, D is any point on BC . Prove that $AC^2 = AD^2 + BC^2 - CD^2$.

****Case-Based Question:****

Read the following and answer:

"In a right-angled triangle, the hypotenuse is 25 cm and one side is 7 cm."

- (i) Find the length of the other side.
- (ii) State the theorem used.
- (iii) What is the area of the triangle?
- (iv) What is the perimeter of the triangle?

CHAPTER 8: CIRCLES

1. Tangent to a Circle

****Definition:****

A tangent to a circle is a line that touches the circle at exactly one point.

****Point of Contact:****

The point where the tangent touches the circle.

****Secant:****

A line that intersects the circle at two points.

****Important Properties:****

1. The tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. The lengths of tangents drawn from an external point to a circle are equal.

2. Theorem: Tangent is Perpendicular to Radius

****Statement:****

The tangent at any point of a circle is perpendicular to the radius through the point of contact.

****Proof:****

Let O be the centre of the circle, and AB be a tangent at point P.

****Steps:****

1. Take a point Q on AB other than P.
2. Since Q is outside the circle, $OQ > OP$.
3. Since P is the point of contact, OP is the shortest distance from O to the line AB.
4. The shortest distance from a point to a line is the perpendicular distance.
5. Therefore, $OP \perp AB$.

3. Theorem: Lengths of Tangents from an External Point

****Statement:****

The lengths of tangents drawn from an external point to a circle are equal.

****Proof:****

Let O be the centre of the circle, and P be an external point. PA and PB are two tangents to the circle at points A and B respectively.

****Steps:****

1. Join OA, OB, and OP.
2. In $\triangle OAP$ and $\triangle OBP$:
 - $OA = OB$ (radii)
 - $OP = OP$ (common)
 - $\angle OAP = \angle OBP = 90^\circ$ (tangent is perpendicular to radius)
3. Therefore, $\triangle OAP \cong \triangle OBP$ (RHS congruence)
4. $PA = PB$ (CPCT)

4. Number of Tangents from a Point

- **Point inside the circle:** No tangent.
- **Point on the circle:** Exactly one tangent.
- **Point outside the circle:** Exactly two tangents.

5. Important Results

1. **Tangent-Secant Theorem:**

If a tangent and a secant are drawn from an external point to a circle, then the square of the tangent is equal to the product of the entire secant and its external part.

2. **Cyclic Quadrilateral:**

The sum of the opposite angles of a cyclic quadrilateral is 180° .

IMPORTANT QUESTIONS

Very Short Answer (1 mark):

1. Define tangent to a circle.
2. What is the point of contact?
3. How many tangents can be drawn from an external point to a circle?
4. What is the angle between the tangent and the radius at the point of contact?
5. Write the theorem about the lengths of tangents from an external point.

Short Answer (2-3 marks):

1. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Prove that the lengths of tangents drawn from an external point to a circle are equal.
3. If two tangents are drawn from an external point to a circle, what can you say about their lengths?
4. Find the length of a tangent from a point 5 cm away from the centre of a circle of radius 3 cm.

Long Answer (4-5 marks):

1. Prove the theorem: The tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Prove the theorem: The lengths of tangents drawn from an external point to a circle are equal.

3. From a point P, two tangents PA and PB are drawn to a circle with centre O. If $OP = 13$ cm and radius = 5 cm, find the length of PA.

****Case-Based Question:****

Read the following and answer:

"From a point P outside a circle, two tangents PA and PB are drawn to the circle with centre O."

- (i) What is the relationship between PA and PB?
- (ii) What is the angle between PA and OA?
- (iii) If $\angle AOP = 60^\circ$, what is $\angle AOB$?
- (iv) If $OP = 10$ cm and radius = 6 cm, find PA.

UNIT 5: TRIGONOMETRY

CHAPTER 9: INTRODUCTION TO TRIGONOMETRY

1. Trigonometric Ratios

****Definition:****

In a right-angled triangle ABC, with $\angle B = 90^\circ$:

- $\sin A = \text{Opposite side} / \text{Hypotenuse} = BC/AC$
- $\cos A = \text{Adjacent side} / \text{Hypotenuse} = AB/AC$
- $\tan A = \text{Opposite side} / \text{Adjacent side} = BC/AB$
- $\operatorname{cosec} A = 1/\sin A = AC/BC$
- $\sec A = 1/\cos A = AC/AB$
- $\cot A = 1/\tan A = AB/BC$

****Example:****

In a right-angled triangle, if $\sin \theta = 3/5$, find $\cos \theta$ and $\tan \theta$.

- $\sin \theta = 3/5 = \text{Opposite/Hypotenuse}$
- Let Opposite = 3k, Hypotenuse = 5k
- Adjacent = $\sqrt{(5k)^2 - (3k)^2} = \sqrt{16k^2} = 4k$
- $\cos \theta = 4/5$
- $\tan \theta = 3/4$

2. Trigonometric Ratios of Standard Angles

Angle	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	Not defined
cosec	Not defined	2	√2	2/√3	1
sec	1	2/√3	√2	2	Not defined
cot	Not defined	√3	1	1/√3	0

3. Trigonometric Identities

****Fundamental Identities:****

1. $\sin^2 A + \cos^2 A = 1$
2. $1 + \tan^2 A = \sec^2 A$
3. $1 + \cot^2 A = \operatorname{cosec}^2 A$

****Proof of $\sin^2 A + \cos^2 A = 1$:**

- $\sin A = BC/AC$, $\cos A = AB/AC$

$$- \sin^2 A + \cos^2 A = BC^2/AC^2 + AB^2/AC^2 = (BC^2 + AB^2)/AC^2 = AC^2/AC^2 = 1$$

****Other Forms:****

- $\sin^2 A = 1 - \cos^2 A$
- $\cos^2 A = 1 - \sin^2 A$
- $\sec^2 A = 1 + \tan^2 A$
- $\operatorname{cosec}^2 A = 1 + \cot^2 A$

****Example:****

Prove that:

$$- (\sin A + \cos A)^2 = 1 + 2 \sin A \cos A$$

$$- \text{LHS} = \sin^2 A + \cos^2 A + 2 \sin A \cos A = 1 + 2 \sin A \cos A = \text{RHS}$$

4. Complementary Angles

****Relationships:****

- $\sin(90^\circ - \theta) = \cos \theta$
- $\cos(90^\circ - \theta) = \sin \theta$
- $\tan(90^\circ - \theta) = \cot \theta$
- $\cot(90^\circ - \theta) = \tan \theta$

- $\sec(90^\circ - \theta) = \operatorname{cosec} \theta$
- $\operatorname{cosec}(90^\circ - \theta) = \sec \theta$

****Example:****

- $\sin 60^\circ = \cos 30^\circ$
- $\tan 45^\circ = \cot 45^\circ$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the three fundamental trigonometric identities.
2. What is the value of $\sin 0^\circ$?
3. What is the value of $\tan 45^\circ$?
4. What is the value of $\cos 60^\circ$?
5. What is $\sin 90^\circ \times \cos 0^\circ$?

****Short Answer (2-3 marks):****

1. Find the value of $\sin 30^\circ \times \cos 60^\circ + \tan 45^\circ$.
2. If $\sin \theta = 3/5$, find $\cos \theta$ and $\tan \theta$.
3. Prove that: $(\sin^2 \theta + \cos^2 \theta) + (1 + \tan^2 \theta) = \sec^2 \theta$.
4. Find the value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

****Long Answer (4-5 marks):****

1. Prove that: $(\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2$.
2. Prove that: $(1 + \tan^2 A) + (1 + \cot^2 A) = \sec^2 A \times \operatorname{cosec}^2 A$.
3. Evaluate: $(\sin^2 30^\circ + \cos^2 45^\circ + \tan^2 60^\circ) / (\sec^2 30^\circ + \cot^2 45^\circ)$

****Case-Based Question:****

Read the following and answer:

"In a right-angled triangle, the sides are in the ratio 3 : 4 : 5."

- (i) Find $\sin A$.
- (ii) Find $\cos A$.
- (iii) Find $\tan A$.
- (iv) Verify $\sin^2 A + \cos^2 A = 1$.

CHAPTER 10: APPLICATIONS OF TRIGONOMETRY

1. Angle of Elevation

****Definition:****

The angle formed by the line of sight with the horizontal when the object is above the observer.

2. Angle of Depression

****Definition:****

The angle formed by the line of sight with the horizontal when the object is below the observer.

3. Heights and Distances

****Applications:****

- Finding the height of a tower, building, mountain, or tree.
- Finding the distance of an object from the observer.
- Finding the distance between two objects.

****Steps to Solve Problems:****

1. Draw a figure representing the problem.
2. Mark all the given angles and distances.
3. Identify the right-angled triangles.
4. Use appropriate trigonometric ratios to find the unknown.

****Note:****

- Problems should not involve more than two right-angled triangles.
- Angles of elevation/depression are only 30° , 45° , and 60° .

****Example:****

A tower stands vertically on the ground. From a point on the ground, the angle of elevation of the top of the tower is 60° . If the distance from the point to the foot of the tower is 10 m, find the height of the tower.

Let the height of the tower be h .

- $\tan 60^\circ = h/10$
- $\sqrt{3} = h/10$
- $h = 10\sqrt{3}$ m

****Example:****

The angle of elevation of the top of a building from a point on the ground is 30° . If the height of the building is 20 m, find the distance from the point to the foot of the building.

Let the distance be d .

$$- \tan 30^\circ = 20/d$$

$$- 1/\sqrt{3} = 20/d$$

$$- d = 20\sqrt{3} \text{ m}$$

****Example:****

A ladder of length 10 m is placed against a wall. The ladder makes an angle of 60° with the ground. Find the height of the wall and the distance of the foot of the ladder from the wall.

Let height = h , distance = d .

$$- \sin 60^\circ = h/10 \rightarrow \sqrt{3}/2 = h/10 \rightarrow h = 5\sqrt{3} \text{ m}$$

$$- \cos 60^\circ = d/10 \rightarrow 1/2 = d/10 \rightarrow d = 5 \text{ m}$$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Define angle of elevation.
2. Define angle of depression.
3. What is the angle of elevation of the top of a tower if its height is equal to the distance from the observer?
4. What trigonometric ratio would you use to find the height of a tower if the angle of elevation and distance are given?

****Short Answer (2-3 marks):****

1. A tree casts a shadow of 10 m when the angle of elevation of the sun is 45° . Find the height of the tree.
2. The angle of elevation of the top of a tower from a point on the ground is 30° . If the distance from the point to the foot of the tower is 20 m, find the height of the tower.
3. The angle of depression of a car from the top of a building 20 m high is 45° . Find the distance of the car from the foot of the building.

****Long Answer (4-5 marks):****

1. A man standing at a distance of 30 m from the foot of a tower observes the angle of elevation of the top of the tower as 30° . Find the height of the tower. After some time, he moves towards the tower and observes the angle of elevation as 60° . How far has he moved?
2. The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the height of the tower is 30 m, find the height of the building and the distance between them.

****Case-Based Question:****

Read the following and answer:

"A girl of height 1.5 m stands at a distance of 20 m from a lamp post. The angle of elevation of the top of the lamp post from her eye is 60° ."

- (i) Find the height of the lamp post.
- (ii) What is the angle of elevation?
- (iii) What trigonometric ratio is used here?
- (iv) What would be the height if the angle was 30° ?

UNIT 6: MENSURATION

CHAPTER 11: AREAS RELATED TO CIRCLES

1. Circumference and Area of a Circle

****Circumference:****

- $C = 2\pi r$
- Where r is the radius.

****Area:****

- $A = \pi r^2$

****Example:****

Find the circumference and area of a circle of radius 7 cm.

- $C = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$
- $A = \frac{22}{7} \times 7^2 = 154 \text{ cm}^2$

2. Area of a Sector

****Sector:****

The region bounded by two radii and the corresponding arc.

****Angle of Sector:****

- θ = angle subtended at the centre.

****Area of Sector:****

- $\text{Area} = \left(\frac{\theta}{360^\circ}\right) \times \pi r^2$

****Length of Arc:****

- Arc length = $(\theta/360^\circ) \times 2\pi r$

****Example:****

Find the area of a sector of a circle of radius 6 cm and angle 60° .

- Area = $(60/360) \times \pi \times 6^2 = 1/6 \times 36\pi = 6\pi \text{ cm}^2$

3. Area of a Segment

****Segment:****

The region bounded by a chord and the corresponding arc.

****Area of Minor Segment:****

- Area = Area of sector - Area of triangle formed by the chord and radii.

****Area of Major Segment:****

- Area = Area of circle - Area of minor segment.

****Example:****

Find the area of a minor segment of a circle of radius 6 cm and angle 60° .

- Area of sector = $(60/360) \times \pi \times 6^2 = 6\pi \text{ cm}^2$

- Area of triangle = $1/2 \times r^2 \times \sin \theta = 1/2 \times 36 \times \sin 60^\circ = 18 \times \sqrt{3}/2 = 9\sqrt{3} \text{ cm}^2$

- Area of segment = $6\pi - 9\sqrt{3} \text{ cm}^2$

****Note:****

- Problems are restricted to central angles of 60° , 90° , and 120° .

4. Important Results

****1. Area of the ring:****

- Area = $\pi(R^2 - r^2)$, where R is the outer radius and r is the inner radius.

****2. Area of the sector with perimeter:****

- Perimeter of sector = $2r + (\theta/360^\circ) \times 2\pi r$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the formula for the area of a circle.
2. Write the formula for the area of a sector.
3. What is the area of a segment?
4. What is the circumference of a circle of radius 14 cm?
5. What is the area of a circle of radius 7 cm?

****Short Answer (2-3 marks):****

1. Find the area of a sector of a circle of radius 10 cm and angle 45° .
2. Find the area of a minor segment of a circle of radius 6 cm and angle 60° .
3. The area of a sector is $77/3 \text{ cm}^2$ and the radius is 7 cm. Find the angle of the sector.
4. Find the area of a ring whose outer radius is 10 cm and inner radius is 6 cm.

****Long Answer (4-5 marks):****

1. Find the area of the shaded region in a circle of radius 14 cm, where a sector of angle 90° is cut out.
2. A chord of a circle of radius 10 cm subtends an angle of 60° at the centre. Find the area of the corresponding major segment.
3. Find the perimeter and area of a sector of a circle of radius 12 cm and angle 60° .

****Case-Based Question:****

Read the following and answer:

"A circle of radius 14 cm has a sector of angle 60° ."

- (i) Find the area of the sector.
- (ii) Find the area of the triangle formed by the radii and the chord.
- (iii) Find the area of the minor segment.
- (iv) What is the circumference of the circle?

CHAPTER 12: SURFACE AREAS AND VOLUMES

1. Surface Areas and Volumes of Basic Solids

****A. Cube:****

- ****Lateral Surface Area (LSA):**** $4a^2$
- ****Total Surface Area (TSA):**** $6a^2$
- ****Volume:**** a^3

****B. Cuboid:****

- ****Lateral Surface Area (LSA):**** $2h(l + b)$

- **Total Surface Area (TSA):** $2(lb + bh + hl)$
- **Volume:** $l \times b \times h$

C. Cylinder:

- **Curved Surface Area (CSA):** $2\pi rh$
- **Total Surface Area (TSA):** $2\pi r(r + h)$
- **Volume:** $\pi r^2 h$

D. Cone:

- **Curved Surface Area (CSA):** πrl
- **Total Surface Area (TSA):** $\pi r(r + l)$
- **Volume:** $\frac{1}{3} \pi r^2 h$
- **Slant Height (l):** $\sqrt{r^2 + h^2}$

E. Sphere:

- **Curved Surface Area (CSA):** $4\pi r^2$
- **Total Surface Area (TSA):** $4\pi r^2$
- **Volume:** $\frac{4}{3} \pi r^3$

F. Hemisphere:

- **Curved Surface Area (CSA):** $2\pi r^2$
- **Total Surface Area (TSA):** $3\pi r^2$
- **Volume:** $\frac{2}{3} \pi r^3$

2. Combination of Solids

Types of Combinations:

1. **Frustum of a Cone:**

- $CSA = \pi l(R + r)$, where $l = \sqrt{(R - r)^2 + h^2}$
- $TSA = \pi(R^2 + r^2 + Rl + rl)$
- $Volume = \frac{1}{3} \pi h(R^2 + r^2 + Rr)$

2. **Composite Solids:**

- A solid made up of two or more basic solids.
- Find the surface area and volume by adding or subtracting the areas/volumes of the individual parts.

Example:

Find the surface area of a cone mounted on a cylinder.

- $TSA = CSA \text{ of cone} + CSA \text{ of cylinder} + \text{Area of the base}$

Example:

Find the volume of a sphere attached to a cylinder.

- Volume = Volume of sphere + Volume of cylinder

****Important Points:****

- When two solids are joined, the surface area of the joined part is not counted.
- The total surface area of the combined solid is the sum of the surface areas of the individual solids minus the area of the common part twice.

3. Conversion of Solid from One Shape to Another

****Example:****

A solid sphere is melted and recast into a cylinder. Find the dimensions of the cylinder.

- Volume of sphere = Volume of cylinder

$$- \frac{4}{3} \pi R^3 = \pi r^2 h$$

****Example:****

A cone is melted and recast into a sphere. Find the radius of the sphere.

- Volume of cone = Volume of sphere

$$- \frac{1}{3} \pi r^2 h = \frac{4}{3} \pi R^3$$

4. Problems on Volume and Capacity

****Example:****

Find the number of small spheres that can be made from a larger sphere.

- Number of small spheres = Volume of large sphere / Volume of small sphere

****Example:****

How much water can a cylinder hold?

- Capacity = Volume of cylinder

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the formula for the volume of a cylinder.

2. Write the formula for the curved surface area of a cone.
3. Write the formula for the volume of a sphere.
4. Write the formula for the total surface area of a cuboid.
5. What is the formula for the slant height of a cone?

****Short Answer (2-3 marks):****

1. Find the volume of a cube of side 5 cm.
2. Find the curved surface area of a cylinder of radius 7 cm and height 10 cm.
3. Find the volume of a cone of radius 3 cm and height 4 cm.
4. Find the surface area of a sphere of radius 7 cm.

****Long Answer (4-5 marks):****

1. A solid cylinder of radius 7 cm and height 10 cm is melted and recast into a cone of radius 7 cm. Find the height of the cone.
2. Find the total surface area of a cone of radius 3 cm and height 4 cm.
3. A hemispherical tank of radius 7 m is full of water. It is emptied into a cylindrical tank of radius 7 m. Find the height of the water in the cylindrical tank.

****Case-Based Question:****

Read the following and answer:

"A cone of radius 3 cm and height 4 cm is melted and recast into a sphere."

- (i) Find the slant height of the cone.
- (ii) Find the volume of the cone.
- (iii) Find the radius of the sphere.
- (iv) What is the surface area of the sphere?

UNIT 7: STATISTICS AND PROBABILITY

CHAPTER 13: STATISTICS

1. Mean of Grouped Data

****A. Direct Method:****

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

****Example:****

Find the mean of the following data:

Class	0-10	10-20	20-30	30-40	40-50
-----	-----	-----	-----	-----	-----
f	5	10	15	12	8

- x_i (midpoints): 5, 15, 25, 35, 45

- $\Sigma f = 5 + 10 + 15 + 12 + 8 = 50$

- $\Sigma f x = 5 \times 5 + 10 \times 15 + 15 \times 25 + 12 \times 35 + 8 \times 45 = 25 + 150 + 375 + 420 + 360 = 1330$

- Mean = $1330/50 = 26.6$

****B. Assumed Mean Method:****

Mean = $a + (\Sigma f d_i / \Sigma f_i)$, where $d_i = x_i - a$

****C. Step Deviation Method:****

Mean = $a + h \times (\Sigma f u_i / \Sigma f_i)$, where $u_i = (x_i - a)/h$

****Example:****

Find the mean using the step deviation method for the above data.

- $a = 25$, $h = 10$

- $u_i = (x_i - 25)/10$: -2, -1, 0, 1, 2

- $\Sigma f = 50$

- $\Sigma f u = 5(-2) + 10(-1) + 15(0) + 12(1) + 8(2) = -10 - 10 + 0 + 12 + 16 = 8$

- Mean = $25 + 10 \times (8/50) = 25 + 1.6 = 26.6$

2. Median of Grouped Data

****Steps:****

1. Find the cumulative frequency (cf).

2. Identify the median class: the class where $N/2$ lies ($N = \Sigma f$).

3. Median = $l + [(N/2 - cf)/f] \times h$

- l = lower limit of median class

- cf = cumulative frequency of the class preceding the median class

- f = frequency of the median class

- h = class size

****Example:****

Find the median of the data:

Class	0-10	10-20	20-30	30-40	40-50
f	5	10	15	12	8

- $N = 50$, $N/2 = 25$
- cf: 5, 15, 30, 42, 50
- Median class: 20-30 (since cf ≥ 25)
- $l = 20$, cf = 15, $f = 15$, $h = 10$
- Median = $20 + [(25 - 15)/15] \times 10 = 20 + (10/15) \times 10 = 20 + 6.67 = 26.67$

3. Mode of Grouped Data

****Formula:****

$$\text{Mode} = l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$$

Where:

- l = lower limit of the modal class (highest frequency)
- f_1 = frequency of the modal class
- f_0 = frequency of the class preceding the modal class
- f_2 = frequency of the class succeeding the modal class
- h = class size

****Example:****

Find the mode of the data:

Class	0-10	10-20	20-30	30-40	40-50
f	5	10	15	12	8

- Modal class: 20-30 (highest frequency = 15)
- $l = 20$, $f_1 = 15$, $f_0 = 10$, $f_2 = 12$, $h = 10$
- Mode = $20 + [(15 - 10)/(2 \times 15 - 10 - 12)] \times 10$
- = $20 + [5/(30 - 22)] \times 10 = 20 + (5/8) \times 10 = 20 + 6.25 = 26.25$

4. Relationship between Mean, Median, and Mode

****Empirical Formula:****

- Mode = 3 Median - 2 Mean

****Example:****

If Mean = 30 and Median = 28, find Mode.

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean}) = 3(28) - 2(30) = 84 - 60 = 24$$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. Write the formula for the mean of grouped data.
2. Write the formula for the median of grouped data.
3. Write the formula for the mode of grouped data.
4. What is the empirical relationship between mean, median, and mode?
5. What is the modal class?

****Short Answer (2-3 marks):****

1. Find the mean of the following data using the direct method:
- 5, 10, 15, 20, 25 with frequencies 3, 5, 8, 4, 2
2. Find the median of the data:
- 10, 20, 30, 40, 50 with frequencies 2, 4, 6, 3, 1
3. Find the mode of the data:
- 10, 20, 30, 40, 50 with frequencies 5, 10, 15, 8, 2

****Long Answer (4-5 marks):****

1. Find the mean, median, and mode of the following data:

Class	0-10	10-20	20-30	30-40	40-50
f	4	6	10	8	2

2. The following table shows the marks obtained by students. Find the mean, median, and mode:

Marks	0-10	10-20	20-30	30-40	40-50
Students	5	8	12	6	4

****Case-Based Question:****

Read the following and answer:

"The marks of 30 students in a test are given below:

Marks	0-10	10-20	20-30	30-40	40-50
Students	4	6	8	7	5

- (i) Find the mean of the data.
- (ii) Find the median of the data.
- (iii) Find the mode of the data.
- (iv) Verify the empirical relationship.

CHAPTER 14: PROBABILITY

1. Classical Definition of Probability

Definition:

If an experiment has n equally likely outcomes and m of them are favourable to an event E , then:

- $P(E) = m/n = \text{Number of favourable outcomes} / \text{Total number of possible outcomes}$

Properties:

- $0 \leq P(E) \leq 1$

- $P(E) + P(\text{not } E) = 1$

Note:

- The probability of an impossible event is 0.
- The probability of a certain event is 1.

2. Experimental Probability (Empirical Probability)

Definition:

If an experiment is repeated n times and an event occurs m times, then:

- $P(E) = m/n$

Note:

- Experimental probability is based on actual observations.
- Classical probability is based on theoretical assumptions.

3. Simple Problems on Probability

Example 1:

A coin is tossed once. Find the probability of getting a head.

- Total outcomes = 2 (H, T)
- Favourable outcomes = 1 (H)
- $P(H) = 1/2$

****Example 2:****

A die is thrown once. Find the probability of getting an even number.

- Total outcomes = 6 (1, 2, 3, 4, 5, 6)
- Favourable outcomes = 3 (2, 4, 6)
- $P(\text{even}) = 3/6 = 1/2$

****Example 3:****

A bag contains 5 red and 3 blue balls. A ball is drawn at random. Find the probability that it is red.

- Total outcomes = 8
- Favourable outcomes = 5
- $P(\text{red}) = 5/8$

****Example 4:****

Two coins are tossed simultaneously. Find the probability of getting at least one head.

- Total outcomes = 4 (HH, HT, TH, TT)
- Favourable outcomes = 3 (HH, HT, TH)
- $P(\text{at least one head}) = 3/4$

4. Important Results

****1. Sum of Probabilities:****

- $P(E) + P(\text{not } E) = 1$
- $P(\text{not } E) = 1 - P(E)$

****2. Probability of an Event:****

- $0 \leq P(E) \leq 1$

****3. Mutually Exclusive Events:****

- If two events cannot occur together, they are mutually exclusive.
- $P(A \cup B) = P(A) + P(B)$

****4. Complementary Events:****

- If E is an event, then not E is its complement.

- $P(E) + P(\text{not } E) = 1$

IMPORTANT QUESTIONS

****Very Short Answer (1 mark):****

1. What is the probability of an impossible event?
2. What is the probability of a certain event?
3. What is the sum of the probabilities of an event and its complement?
4. If a coin is tossed twice, what is the total number of outcomes?
5. What is the probability of getting a head when a coin is tossed?

****Short Answer (2-3 marks):****

1. A bag contains 4 red, 3 blue, and 5 green balls. A ball is drawn at random. Find the probability that it is:
(a) red (b) green (c) not blue
2. Two dice are thrown simultaneously. Find the probability of getting:
(a) a sum of 7 (b) a sum of 10
3. A card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting:
(a) a king (b) a red card (c) a face card

****Long Answer (4-5 marks):****

1. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting:
(a) a black card (b) a queen (c) a jack of hearts (d) a diamond
2. Two coins are tossed simultaneously. Find the probability of getting:
(a) two heads (b) one head and one tail (c) at least one head (d) no head
3. A bag contains 5 red, 6 blue, and 7 green balls. Three balls are drawn at random. Find the probability that they are of the same colour.

****Case-Based Question:****

Read the following and answer:

"A box contains 10 red balls, 15 blue balls, and 25 green balls. A ball is drawn at random."

- (i) What is the total number of outcomes?
- (ii) What is the probability of getting a red ball?
- (iii) What is the probability of getting a blue ball?
- (iv) What is the probability of getting a green ball?
- (v) What is the sum of these probabilities?

IMPORTANT FORMULAE

****Number Systems:****

- $\text{HCF} \times \text{LCM} = a \times b$

****Algebra:****

- For quadratic polynomial $ax^2 + bx + c$ with zeroes α and β :

- $\alpha + \beta = -b/a$

- $\alpha \times \beta = c/a$

- $(x + y)^2 = x^2 + 2xy + y^2$

- $(x - y)^2 = x^2 - 2xy + y^2$

- $(x + y)(x - y) = x^2 - y^2$

- $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$

- $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

- $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$

- $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

- $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

- $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

****Linear Equations:****

- For equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$:

- Unique solution: $a_1/a_2 \neq b_1/b_2$

- Infinite solutions: $a_1/a_2 = b_1/b_2 = c_1/c_2$

- No solution: $a_1/a_2 = b_1/b_2 \neq c_1/c_2$

****Quadratic Equations:****

- $ax^2 + bx + c = 0$

- $x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$

- Discriminant: $D = b^2 - 4ac$

- $D > 0$: Two distinct real roots

- $D = 0$: Equal roots

- $D < 0$: No real roots

****Arithmetic Progressions:****

- $a_n = a + (n - 1)d$

- $S_n = n/2[2a + (n - 1)d]$

- $S_n = n/2(a + l)$

****Coordinate Geometry:****

- Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

- Section formula (internal division): $(m_1x_2 + m_2x_1)/(m_1 + m_2), (m_1y_2 + m_2y_1)/(m_1 + m_2)$

- Midpoint formula: $((x_1 + x_2)/2, (y_1 + y_2)/2)$

- Area of triangle: $1/2 |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

****Geometry:****

- BPT: $AD/DB = AE/EC$

- Pythagoras Theorem: $AC^2 = AB^2 + BC^2$
- Tangent is perpendicular to radius.
- Lengths of tangents from an external point are equal.

****Trigonometry:****

- $\sin \theta = \text{Opposite/Hypotenuse}$
- $\cos \theta = \text{Adjacent/Hypotenuse}$
- $\tan \theta = \text{Opposite/Adjacent}$
- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \text{cosec}^2 \theta$
- $\sin(90^\circ - \theta) = \cos \theta$
- $\cos(90^\circ - \theta) = \sin \theta$
- $\tan(90^\circ - \theta) = \cot \theta$

****Mensuration:****

- Cube: TSA = $6a^2$, Volume = a^3
- Cuboid: TSA = $2(lb + bh + hl)$, Volume = $l \times b \times h$
- Cylinder: CSA = $2\pi rh$, TSA = $2\pi r(r + h)$, Volume = $\pi r^2 h$
- Cone: CSA = πrl , TSA = $\pi r(r + l)$, Volume = $\frac{1}{3} \pi r^2 h$, $l = \sqrt{r^2 + h^2}$
- Sphere: SA = $4\pi r^2$, Volume = $\frac{4}{3} \pi r^3$
- Hemisphere: CSA = $2\pi r^2$, TSA = $3\pi r^2$, Volume = $\frac{2}{3} \pi r^3$
- Sector: Area = $(\theta/360^\circ) \times \pi r^2$, Arc length = $(\theta/360^\circ) \times 2\pi r$
- Segment: Area = Sector area - Triangle area
- Frustum: CSA = $\pi l(R + r)$, TSA = $\pi(R^2 + r^2 + Rl + rl)$, Volume = $\frac{1}{3} \pi h(R^2 + r^2 + Rr)$

****Statistics:****

- Mean (direct) = $\sum f_i x_i / \sum f_i$
- Mean (assumed mean) = $a + (\sum f_i d_i / \sum f_i)$
- Mean (step deviation) = $a + h \times (\sum f_i u_i / \sum f_i)$
- Median = $l + [(N/2 - cf)/f] \times h$
- Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$
- Mode = 3 Median - 2 Mean

****Probability:****

- $P(E) = \text{Number of favourable outcomes} / \text{Total number of possible outcomes}$
- $P(E) + P(\text{not } E) = 1$
- $0 \leq P(E) \leq 1$
